

...sets the line colour to red and fills the inside of the square with yellow colour. We will discuss fillstyle and the other related parameters later. The line of code to draw the circle is:

```
\pscircle[linecolor=blue,fillstyle=solid,fillcolor=green](2,2){1}
```

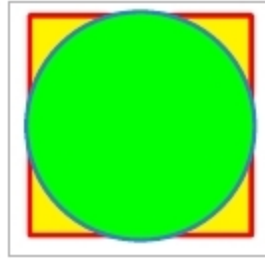


Figure 3: Drawing a circle and a square using PSTricks

This line draws a circle of radius of one unit with the centre at point (2,2). The code fragment:

```
[linecolor=blue,fillstyle=solid,fillcolor=green]
```

...draws the circle in blue colour and fills its interior with green colour. Since the diameter of the circle is two units and the sides of the square are also two units, the circle is inscribed inside the square. On execution with Texmaker or in terminal using commands, the file called *pstricks3.tex* will produce the output shown in Figure 3.

Now it is time for a simple challenge. Can you modify the file *pstricks3.tex* to obtain the image shown in Figure 4? All you need to do is make one or two minor changes in the file *pstricks3.tex*. The first thing to do is increase the radius of the green circle. What is the required radius? Well, here we have a small mathematical problem. The diameter of the required circle is $\sqrt{2}$ units because the sides of the square are of size two units. Since $\sqrt{2}$ is an irrational number, we can't get its exact value. But we have all encountered this problem during our school days — 'inscribe a square inside a circle'. How did we do it back then? Well, there are some Euclidean constructs that give a good approximation for the value of $\sqrt{2}$. Let us also proceed in the same manner; a good approximation for $\sqrt{2}$ is 1.414.

So, the question is this — if we replace the line of code:

```
\pscircle[linecolor=blue,fillstyle=solid,fillcolor=green](2,2){1}
```

...in *pstricks3.tex* with the line of code:

```
\pscircle[linecolor=blue,fillstyle=solid,fillcolor=green](2,2){1.414}
```

...are we going to get the required image as output? The answer is, 'No!' If you do that modification alone, you will see a circle of radius $\sqrt{2}$ drawn in blue colour filled with green colour. Why?

We encountered the same problem when we discussed PGF/TikZ. Both PGF/TikZ and PSTricks draw image components in the order in which the corresponding code appears in the script. Here, we have first drawn a square of sides 2cm and then we have drawn a circle of radius $\sqrt{2}$ over it; hence, the square is not visible. So, the second change to make is to swap the order in which the square and the circle are drawn. Then the required image will be obtained in the output

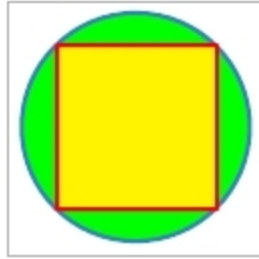


Figure 4: Can you draw this image?

file. The modified script *pstricks4.tex* is also available for download.

Different styles in PSTricks

So far, we have drawn images using simple lines alone. Now it is time for us to draw images in different styles and patterns.

Consider the LaTeX script *pstricks5.tex* shown below, which draws three circles with different fill styles, line styles and newly defined colours.

```
\documentclass{article}
\usepackage{pstricks}
\begin{document}
  \begin{pspicture}(15,15)
    \psset{linewidth=0.05cm}
    \newrgbcolor{col1}{1 1 0.7}
    \newrgbcolor{col2}{0.7 1 1}
    \newrgbcolor{col3}{1 0.7 1}
    \pscircle[linestyle=solid,linecolor=red,fillstyle=vlines*,fillcolor=col1](2,2){1}
    \pscircle[linestyle=dashed,linecolor=green,fillstyle=hlines*,fillcolor=col2](5,2){1}
    \pscircle[linestyle=dotted,linecolor=blue,fillstyle=crosshatch*,fillcolor=col3](8,2){1}
  \end{pspicture}
\end{document}
```

The line of code '`\psset{linewidth=0.05cm}`' sets the line width to 0.05cm. The line of code '`\newrgbcolor{col1}{1 1 0.7}`' defines a new colour with name *col1*. A new colour is specified by providing its RGB components. The values of the RGB components should be between 0 and 1. For example, {0, 0, 0} defines black colour and {1, 1, 1} defines white colour. The next two lines of code define two more new colours named *col2* and *col3*. The line of code '`\pscircle[linestyle=solid,linecolor=red,fillstyle=vlines*,fillcolor=col1](2,2){1}`' draws a circle with the border in red solid line, filled with the colour defined by the variable *col1* and with *fillstyle vlines** filling the inside of the circle by drawing backward slanting lines.

The next line of code draws a circle with a border in a green dashed line, filled with the colour defined by the variable *col2* and with *fillstyle hlines** filling the inside by drawing forward slanting lines. And the following line of code draws a circle with the border in a blue dotted line, filled with the colour defined by the variable *col3* and with *fillstyle crosshatch** filling the inside of the circle by drawing a mesh. On executing the program, *pstricks5.tex* will produce the image shown in Figure 5 as output. Can you modify the code in such a way that the filled colours of the circles are yellow, magenta and cyan? I will give you a hint. Adding red and green gives yellow, adding red and blue gives magenta, and adding green and blue gives cyan. So, what should be the colour codes and where should we make the changes? If you